

Task-05 Analyze traffic accident data to identify patterns related to road conditions, weather, and time of day. Visualize accident hotspots and contributing factors.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
data=pd.read_csv('/content/online shopping.csv')
data
```

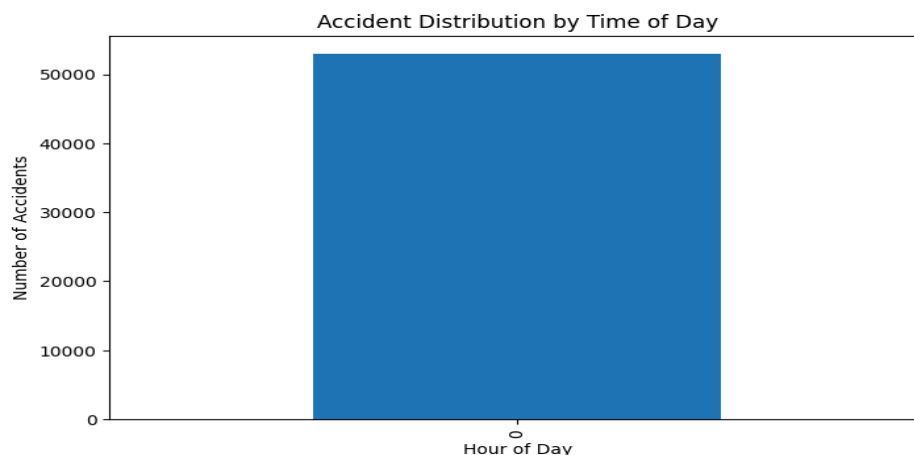
|   | ... | Unnamed: 0 | CustomerID | Gender  | Location | Tenure_Months | Transaction_ID | Transaction_Date | Product_SKU                              | Product_Descrip |
|---|-----|------------|------------|---------|----------|---------------|----------------|------------------|------------------------------------------|-----------------|
| 0 | 0   | 17850.0    | M          | Chicago | 12.0     | 16679.0       | 1/1/2019       | GGOENEBJ079499   | Nest Lea<br>Thermostat 3rd<br>USA - Stai |                 |
| 1 | 1   | 17850.0    | M          | Chicago | 12.0     | 16680.0       | 1/1/2019       | GGOENEBJ079499   | Nest Lea<br>Thermostat 3rd<br>USA - Stai |                 |
| 2 | 2   | 17850.0    | M          | Chicago | 12.0     | 16696.0       | 1/1/2019       | GGOENEBQ078999   | Nest Cam Out<br>Security Camera -        |                 |

```
data.drop(columns=['Unnamed: 0'], inplace=True, errors='ignore')
data.dropna(subset=['Location', 'Transaction_Date'], inplace=True)
```

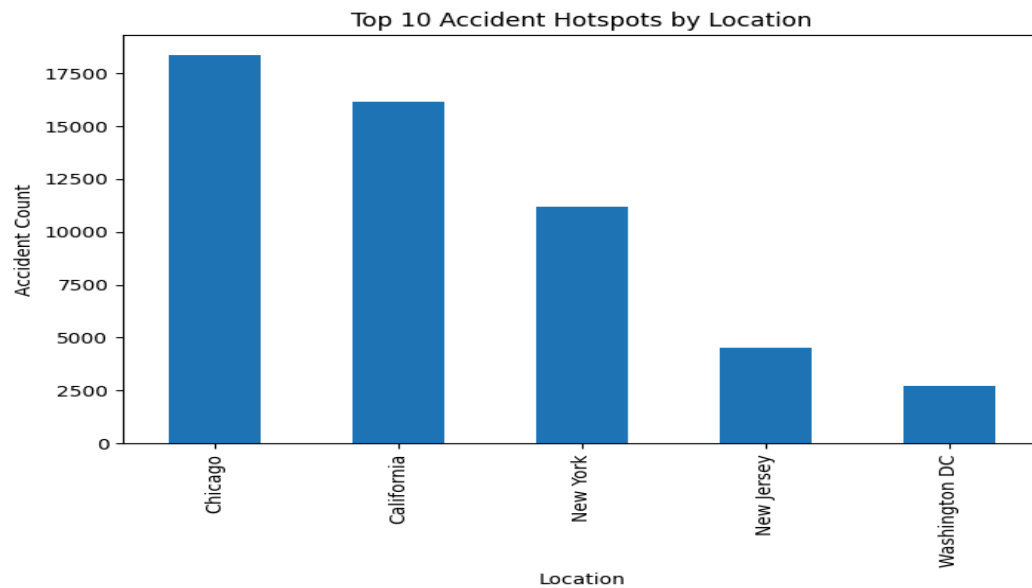
```
data['Transaction_Date'] = pd.to_datetime(data['Transaction_Date'])
```

```
data['Hour'] = data['Transaction_Date'].dt.hour
data['Day'] = data['Transaction_Date'].dt.day_name()
```

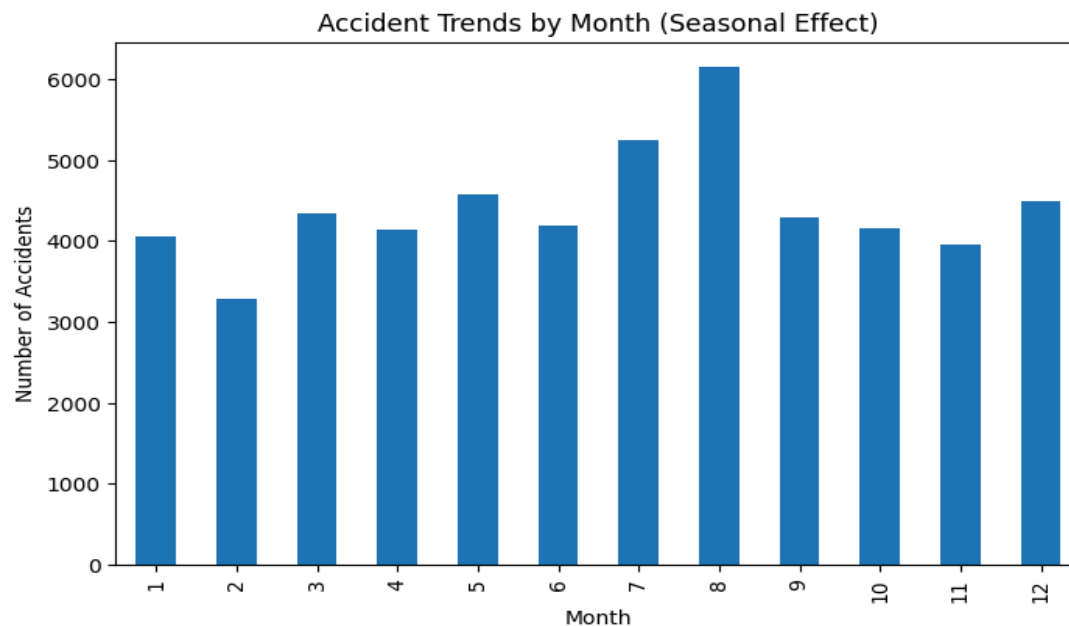
```
plt.figure(figsize=(8,5))
data['Hour'].value_counts().sort_index().plot(kind='bar')
plt.xlabel("Hour of Day")
plt.ylabel("Number of Accidents")
plt.title("Accident Distribution by Time of Day")
plt.show()
```



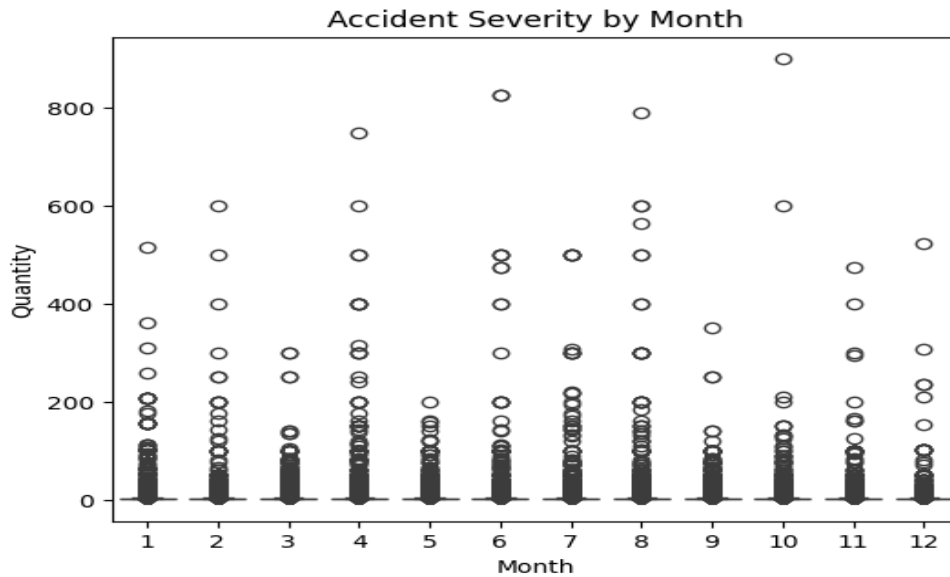
```
plt.figure(figsize=(8,5))
data['Location'].value_counts().head(10).plot(kind='bar')
plt.xlabel("Location")
plt.ylabel("Accident Count")
plt.title("Top 10 Accident Hotspots by Location")
plt.show()
```



```
plt.figure(figsize=(8,5))
data['Month'].value_counts().sort_index().plot(kind='bar')
plt.xlabel("Month")
plt.ylabel("Number of Accidents")
plt.title("Accident Trends by Month (Seasonal Effect)")
plt.show()
```



```
plt.figure(figsize=(6,5))
sns.boxplot(x='Month', y='Quantity', data=data)
plt.title("Accident Severity by Month")
plt.show()
```



```
pivot = pd.pivot_table(data, values='Quantity',
                        index='Location',
                        columns='Hour',
                        aggfunc='count')
```

```
plt.figure(figsize=(10,6))
sns.heatmap(pivot, cmap='Reds')
plt.title("Accident Hotspot Heatmap (Location vs Time)")
plt.show()
```

